

1. Which vein is responsible for venous reflux when the venous refill time normalizes after application of a thigh tourniquet?

- A. Perforating
- B. Femoral
- C. Small saphenous
- ☒ D. Great saphenous

2. After detection of a phasic Doppler signal in the common femoral vein, failure to produce augmentation by thigh compression suggests obstruction in which vein?

- ☒ A. External iliac
- B. Femoral
- C. Popliteal
- D. Great saphenous

3. Which feature is common with an arterial ulcer?

- ☒ A. Dry shiny skin
- B. Shallow with flat margins
- C. Located at the gaiter area
- D. Pain improves with elevation and rest

4. Which landmark is commonly used from the transtemporal approach during transcranial doppler?

- ☒ A. Sphenoid bone
- B. Mastoid bone
- C. Distal posterior cerebral artery
- D. Middle cerebral artery M2 branch

5. Which finding is associated with aliasing, low resistance waveform and increased diastolic flow in a patient status post liver biopsy?

- A. Hematoma
- B. Portal vein occlusion
- ☒ C. Arteriovenous fistula
- D. Hepatic vein thrombosis

6. Where is the abdominal aorta located?

- ☒ A. Retroperitoneum
- B. Peritoneum
- C. Lesser sac
- D. Morison pouch

7. Which side-to-side blood pressure difference is the threshold that may indicate a hemodynamically significant lesion on the side with the lower pressure?

- A. 10 mmHg
- B. 20 mmHg

- C. 30 mmHg
☒ D. 40 mmHg

??

8. In which artery do the majority of peripheral arterial aneurysms occur?

- A. Radial
B. Brachial
☒ C. Popliteal
D. Posterior tibial

9. Which artery is likely to be involved in a patient with thromboangiitis obliterans (Buerger's Disease)?

- A. Profunda
B. Femoral
C. Popliteal
☒ D. Posterior tibial

10. How does peak systolic velocity change in a normal celiac artery following a meal?

- A. Doubles
☒ B. No appreciable increase
C. Decreases at least 50%
D. Equals the aortic velocity

11. Which action increases venous blood flow from the legs toward the heart?

- ☒ A. Descension of the diaphragm
B. Ascension of the diaphragm
☒ C. Increased abdominal pressure
D. External pressure on the inferior vena cava

??

BOTH DESCRIBING
SAME THING?

12. Which condition can cause nonatherosclerotic narrowing of brachiocephalic arteries overwhelmingly in female patients?

- A. Compartment syndrome
B. Raynaud Syndrome
☒ C. Takayasu arteritis
D. Buerger disease

13. Which health factor may create a low-resistance spectral Doppler waveform in the lower extremity of a healthy individual?

- A. Rest
☒ B. Exercise
C. Cold temperature
D. Compression stockings

14. What is the most common cause of transplant failure in an immediate postoperative renal transplant patient?

- A. Drug toxicity
- B. Torsion
- C. Chronic rejection
- ☒ D. Acute tubular necrosis

15. Which flow pattern is consistent with a central arterial occlusion?

- A. High resistive flow in the brachial artery
- ☒ B. Increased diastolic flow in the axillary artery
- C. Pulsatile flow in the common carotid artery
- D. Low resistive flow in the vertebral artery

16. What is usually the first manifestation of chronic arterial occlusive disease of the lower extremities?

- A. Cold feet
- ☒ B. Claudication — *pain from PAD relieved by rest*
- C. Night cramps
- D. Numbness

17. Which condition presents with facial swelling, bilateral upper extremity edema, and dyspnea?

- A. Buerger disease
- B. Thoracic outlet syndrome
- C. Paget von Schroetter disease
- ☒ D. Superior vena cava syndrome

18. Which duplex test would be used to confirm the diagnosis of Paget von Schroetter syndrome?

- A. Lower extremity venous
- ☒ B. Upper extremity venous
- C. Lower extremity arterial
- D. Upper extremity arterial

19. Which characteristic is indicative of lower extremity collateral arteries when compared to native vessels?

- A. Indistinguishable
- B. Larger
- C. Shorter
- ☒ D. More numerous

20. Which spectral Doppler waveform is consistent with normal hepatoportal circulation?

- A. Portal vein showing pulsatile hepatofugal flow
- ☒ B. Portal vein showing pulsatile hepatopetal flow
- C. Hepatic veins showing pulsatile hepatofugal flow
- D. Hepatic veins showing pulsatile hepatopetal flow

21. Which artery demonstrates the lowest resistive index?

- A. Proximal aorta
- B. Superior mesenteric artery in a fasting patient
- ☒ C. Renal artery
- D. Distal aorta

22. What is the most likely finding in a patient presenting with a patient arteriovenous fistula, ipsilateral arm, and hand edema, skin discoloration, and prominent chest wall superficial veins?

- A. Occluded Superficial vein
- B. Steal phenomenon
- C. Inflow artery occlusion
- ☒ D. Central venous obstruction

23. The brachiocephalic vein is formed by the confluence of which two veins?

- ☒ A. Internal jugular and subclavian
- B. Brachial and cephalic
- C. External jugular and innominate
- D. Axillary and basilic

24. When evaluating the renal arteries, which finding in the segmental arteries suggests minimal to stenosis?

- ☒ A. Tardus Parvus
- B. Early systolic peak
- C. Acceleration index less than 3m/s
- D. Acceleration time greater than 70 ms

25. When using photoplethysmography, where should the sensor be placed to document venous insufficiency?

- A. Over a varicosity
- B. On the ulcer
- C. On the lateral calf
- ☒ D. Above the medial malleolus

26. What is a potential cause of primary venous insufficiency?

- A. Previous deep venous thrombosis
- ☒ B. Damaged valves
- C. Persistent outflow obstruction
- D. Congenital absence of valves

27. Venous valves are most numerous in which vein?

- A. Inferior Vena Cava
- B. Popliteal
- C. Femoral

☒ D. Posterior tibial

28. Which vascular complication is most commonly associated with a renal allograft?

- ☒ A. Renal artery stenosis
- B. Renal vein stenosis
- C. Renal vein thrombosis
- D. Renal artery thrombosis

29. Compression from which muscle most frequently causes thoracic outlet syndrome?

- ☒ A. Scalene
- B. Intercostal
- C. Pectoralis minor
- D. Sartorius

30. What is a cause of compartment syndrome?

- A. Gangrene of an extremity
- B. Compression of the popliteal artery by the gastrocnemius muscle?
- C. Severe atherosclerosis of the popliteal artery
- ☒ D. Swelling within the fascial bundles of an extremity

31. Which finding results in a loss of phasicity in the common femoral vein?

- ☒ A. Partial occlusion of the iliac vein
- B. Partial occlusion of the femoral vein
- C. Complete occlusion of the popliteal vein
- D. Complete occlusion of the profunda femoris vein

32. In the upper arm, moving from medial to lateral, veins are normally found in which order?

- A. Cephalic, basilic and brachial
- B. Brachial, basilic and cephalic
- ☒ C. Basilic, brachial and cephalic
- D. Cephalic, brachial and basilic

33. What is the main cause of acute ischemia in the lower extremity?

- A. Atherosclerosis
- B. Trauma
- C. Aneurysm
- ☒ D. Emboli

34. What carotid artery segment is most commonly affected by fibromuscular dysplasia?

- ☒ A. Distal internal
- B. Distal common
- C. Proximal internal
- D. Proximal common

35. What is the most common cause of cerebrovascular stroke?

- A. Middle cerebral artery thromboembolism
- B. Rupture of cerebral aneurysm in Circle of Willis
- ☒ C. Essential hypertension
- D. Brain concussion

36. A continuous Doppler signal in the axillary vein suggests thrombosis in which venous segment?

- ☒ A. Subclavian
- B. Internal jugular
- C. External jugular
- D. Brachial

37. Which technique should be used when evaluating the central veins?

- ☒ A. Compression
- B. Distal analysis
- C. Apply tourniquet
- D. Photoplethysmography

38. Which sonographic finding is normal in a patient after a carotid endarterectomy?

- A. Dissection
- B. Intimal flap
- C. Retained plaque
- ☒ D. Absence of the intima-media stripe

39. Which condition is most often associated with subclavian artery aneurysm?

- ??
- ☒ A. Shoulder pain
 - ☒ B. Digital ischemia
 - ~~C. Venous thrombosis~~
 - ~~D. Cerebral aneurysm~~

40. Which exam should be performed for a patient diagnosed with Hollenhurst plaque?

- A. Lower extremity arterial duplex
- B. Mesenteric artery duplex
- ☒ C. Carotid artery duplex
- D. Abdominal aorta duplex

41. Which finding is seen in the ipsilateral ophthalmic artery, secondary to internal carotid artery occlusion?

- ?
- A. Absent flow
 - B. Flow reversal
 - C. Increase in mean velocity
 - ☒ D. Decrease in mean velocity

42. Which symptoms are associated with intermittent claudication of the lower extremity?

- ☒ A. Leg muscle pain
- B. Restless leg
- C. Calf swelling
- D. Foot numbness

43. The onset to peak seen in arterial waveforms represents which measurement?

- A. Pulsatility index
- B. Resistivity gradient
- ☒ C. Acceleration time
- D. Systolic-diastolic ratio

44. Which measurement is the most dependable tool for assessing the severity of stenotic lesion in a bypass graft in the lower extremity?

- A. Resistive index
- B. Acceleration time
- C. End-diastolic velocity ratio
- ☒ D. Peak systolic velocity ratio

45. In a normal transforaminal transcranial Doppler exam, which artery is insonated at a depth of 60 mm?

- A. Basilar
- ☒ B. Middle cerebral
- C. Posterior cerebral
- D. Vertebral

46. Which peak systolic velocity in an infrainguinal bypass graft indicates a threatened graft?

- ☒ A. 30 cm/s
- B. 50 cm/s
- C. 70 cm/s
- D. 90 cm/s

> 300 cm/s OR < 45 cm/s

47. Which condition is most likely associated with reversed flow in an enlarged left gastric vein on a duplex scan?

- ☒ A. Portal hypertension
- B. Budd-Chiari syndrome
- C. Splenic vein thrombosis
- D. Inferior vena cava thrombosis

48. Which artery is an alternative flow pathway in the presence of severe proximal subclavian artery stenosis?

- A. Axillary
- ☒ B. Vertebral
- C. Common carotid

D. Brachiocephalic

49. Which condition would most likely cause an abnormal volume pulse recording at the high thigh location?

- A. Popliteal artery occlusion
- B. Femoral artery occlusion
- C. Internal Iliac artery stenosis
- ☒ D. Significant aortoiliac disease

50. Which characteristic is consistent with a type II endoleak?

- ☒ A. Blood flow into the aneurysm sac from the distal attachment
- B. Endoleak due to graft porosity
- C. Retrograde flow from the inferior mesenteric artery or lumbar arteries
- D. Endoleak originating from a structural defect of the endograft

51. Which artery is the source of emboli usually involved in left amaurosis fugax?

- A. Left subclavian
- B. Right internal carotid
- ☒ C. Left internal carotid
- D. Left external carotid

52. In a normal transcranial Doppler exam, which demonstrates bilateral flow?

- ??
- ☒ A. Middle cerebral
 - B. Anterior cerebral
 - ☒ C. Terminal internal carotid
 - D. Terminal posterior cerebral

53. Which finding during an endovenous ablation would result in a discontinuation of the procedure?

- ?
- A. Incompetent valve
 - B. Thrombosed varicosity
 - ☒ C. Acute deep vein thrombosis
 - D. Diameter greater than 0.40 cm

54. Which ankle-brachial index is most consistent with claudication?

- A. 0.93
- ☒ B. 0.74
- C. 0.45
- D. 0.18

0.4-0.9 ?

55. Which description is associated with a reversed saphenous vein graft?

- A. Valves are broken or excised prior to surgery
- B. Synthetic graft is sewn to the saphenous vein
- ☒ C. Smallest end of the vein is sewn to the proximal anastomosis

D. Saphenous vein is left in the place and anastomosed to the adjacent arteries

56. Which condition is consistent with a dampened photoplethysmogram waveform?

A. Hyperemia

B. Complete palmar arch

☒ C. Raynaud's disease → *dampened = reduced flow/vasc. constriction*

D. Arteriovenous fistula

57. What can cause a continuous waveform in the common femoral vein?

☒ A. Congestive heart failure

B. Aortic dissection

C. Deep vein thrombosis (DVT) in a calf vein

D. Pregnancy

58. Which value drop in an ankle-brachial index from a prior study suggests progressive graft stenosis?

A. 0.05

B. 0.08

C. 0.12

☒ D. 0.15 → *clinically significant?*

59. What is the characteristic waveform of a normal superior mesenteric artery in a fasting patient?

A. High resistance and high diastolic flow

☒ B. High resistance and low diastolic flow

C. Low resistance and high diastolic flow

D. Low resistance and low diastolic flow

60. Which condition may alter the spectral Doppler waveform causing it to appear abnormal in an otherwise normal carotid artery?

☒ A. Hypertension

B. Systemic lupus

C. Graves disease

D. Aortic insufficiency

61. What is the most likely cause of a carotid artery stent obstruction in the absence of stent fracture or deformation?

A. Intraplaque hemorrhage

☒ B. Intimal hyperplasia

C. Arterial thrombosis

D. Dissection

62. Occlusion of which artery is associated with reversal flow in the ipsilateral external carotid artery?

- A. Common carotid
- ☒ B. Proximal internal carotid
- C. Contralateral external carotid
- D. Vertebral

63. Which condition is characterized by delamination of the intimal and medial layers of the carotid artery walls?

- A. Carotid body tumor
- ☒ B. Carotid dissection
- C. Calcified plaque
- D. Fibromuscular dysplasia

64. Which finding is considered normal after a carotid endarterectomy?

- A. Increased size of the bulb
- ☒ B. Decreased size of the bulb
- C. Increased size of the proximal external carotid artery
- D. Decreased size of the proximal internal carotid artery

65. Which condition is the most common sequela of chronic deep venous thrombosis?

- A. Pulmonary embolism
- ☒ B. Valvular dysfunction
- C. Venous gangrene
- D. Phlegmasia cerulea dolens

66. Which condition is most likely present if upper extremity digit photoplethysmography waveform is diminished with radial artery compression?

- A. Subclavian steal
- B. Raynaud's disease
- C. Incomplete palmar arch
- ☒ D. Proximal radial artery occlusion

67. Which description is associated with normal respiratory variation during evaluation of the upper extremity veins?

- ☒ A. Venous flow increases during inspiration
- B. Venous flow increases during expiration
- C. Venous flow is halted during inspiration
- D. Venous flow is unaffected by respiratory variation

68. Which vein is formed by the confluence of the radial and ulnar veins?

- A. Basilic
- ☒ B. Brachial
- C. Cephalic
- D. Antecubital

69. Which of the following describes waveform amplitude associated with a normal photoplethysmography (PPG) digital cold-water immersion stress test?

- A. Increases when compared to baseline
- ☒ B. Decreases for up to 15 minutes before returning to baseline
- C. Decreases for up to 20 minutes before returning to baseline
- ☒ D. Decreases for less than 10 minutes before returning to baseline

70. How does the Valsalva maneuver typically affect normal venous flow?

- ☒ A. Venous flow is interrupted in both the lower limbs and in the upper limbs
- B. Venous flow decreases in the lower limbs and increases in the upper limbs
- C. Venous flow increases in the lower limbs and decreases in the upper limbs
- D. Venous flow increases in the chest and decreases in the abdomen

71. The subclavian artery becomes the axillary artery at which anatomic landmark?

- A. Cervical foramina
- ☒ B. First rib
- C. Adductor hiatus
- D. Thyrocervical trunk

72. Which characteristic is most likely to appear in a spectral Doppler waveform of a normal common carotid artery?

- A. Plug flow
- ☒ B. Laminar flow
- C. Turbulence
- D. Spectral broadening

73. Which finding suggests chronic renal artery stenosis?

- A. Renal-aortic ratio less than 3.5
- B. Resistive index less than or equal to 0.8
- C. Renal pole-to-pole length less than 8 cm
- ☒ D. Renal artery end-diastolic velocity greater than 55 cm/s

74. The cephalic vein empties blood directly into which vein?

- A. Brachial
- B. Brachiocephalic
- ☒ C. Axillary
- D. Superior vena cava

75. What is the appearance of the waveform in the arterial segment proximal to a normal dialysis access?

- A. High resistance high diastolic flow
- ☒ B. Low resistance high diastolic flow
- C. High resistance low diastolic flow
- D. Low resistance low diastolic flow

76. Which artery in normal anatomy is the third major artery arising from the aortic arch?
- A. Left common carotid
 - B. Brachiocephalic
 - ☒ C. Left subclavian
 - D. Right common carotid
77. Which condition is associated with blue toe syndrome?
- A. Diabetic neuropathy
 - B. Intrinsic compression
 - ☒ C. Aortoiliac occlusive disease
 - D. Inferior vena cava thrombosis
78. What type of aneurysm can be treated with thrombin injection?
- A. Fusiform
 - B. Saccular
 - ☒ C. False
 - D. Mycotic
79. What does a thigh pressure index of 0.9 most likely indicate?
- A. Distal occlusion
 - B. Proximal occlusion
 - C. Distal stenosis
 - ☒ D. Proximal stenosis
80. Which arterial study provides the most anatomic information?
- A. Segmental pressures
 - B. Pulse volume recording
 - ☒ C. Duplex ultrasound
 - D. Photoplethysmography
81. Which description is characteristic of renal vessel length?
- ☒ A. Right renal artery longer than left
 - B. Left renal artery longer than right
 - C. Right renal artery and vein longer than left
 - D. Left renal artery and vein longer than right
82. What has the greatest effect on venous flow in the legs of a normal supine patient?
- A. Variations in cardiac output
 - ☒ B. Diaphragm movement
 - C. Aortic dissection and recoil
 - D. Superior vena cava compression
83. Which critical finding necessitates immediate reporting to the referring physician?
- ☒ A. Small caliber distal internal carotid artery (ICA) with no flow noted

??

stroke ?? ↑

pulmonary embolism?

??

- ☒ B. Dilated, noncompressible common femoral vein
- C. Atrophied, partially compressible basilic vein
- D. Posterior tibial ankle-brachial index of 0.80

84. Which length of time is considered normal for venous valve closure?

- ☒ A. 0 to 0.5 seconds
- B. 0.6 to 0.9 seconds
- C. 1 to 1.5 seconds
- D. 1.6 to 2 seconds

85. Which statement is most accurate regarding exercise treadmill testing?

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- ☒ A. Ankle pressure should not increase after exercise
- ☒ B. Proximal occlusive disease may be demonstrated
- C. Rapid recovery of decreased ankle pressure after exercise suggests nonvascular claudication
- D. Pain with exercise without a decrease in the ankle pressure is diagnostic of vascular claudication

86. Which finding is associated with valvular incompetence of the lower extremity after performing dorsiflexion exercises during plethysmography?

- A. Increased venous refilling time
- B. Increased arterial volume
- ☒ C. Decreased venous refilling time
- D. Decreased arterial volume

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88. Which landmark is commonly used from the transtemporal approach during transcranial Doppler?

- ☒ A. Sphenoid bone
- B. Mastoid bone
- C. Distal posterior cerebral artery
- D. Middle cerebral artery M2 branch

89. Which factor may create a low-resistance spectral Doppler waveform in the lower extremity of a healthy individual?

- A. Rest
- ☒ B. Exercise
- C. Cold temperature

D. Compression stockings

90. Which acoustic window requires a reduction in power intensity in order to follow the ALARA (as low as reasonably achievable) principle during transcranial Doppler examination?

- A. Transtemporal
- ☒ B. Transorbital
- C. Suboccipital
- D. Submandibular

91. How does peak systolic velocity change in a normal celiac artery following a meal?

- A. Doubles
- ☒ B. No appreciable increase
- C. Decreases at least 50%
- D. Equals the aortic velocity

92. A patient's resting ankle-brachial index is 1.45. Which step should be performed next in the evaluation of lower extremity segmental pressures?

- A. Perform cold stress test
- B. Perform treadmill stress test
- ☒ C. Measure toe pressures
- D. Place the cuffs high at calf level

93. Why are the cuffs usually inflated to 65 mmHg during a pulse volume recording test?

- A. To augment arterial flow
- B. To augment venous flow
- C. To minimize arterial flow
- ☒ D. To minimize venous flow

94. What is the normal duration of reflux in the great saphenous vein during an evaluation for chronic venous valvular insufficiency?

- ☒ A. Less than 500 milliseconds
- B. 501-600 milliseconds
- C. 601-700 milliseconds
- D. Greater than 701 milliseconds

?

95. Which anomaly is most common in the upper extremity arteries?

- A. Absent radial
- B. Duplicated brachial
- C. Ulnar origin of axillary
- ☒ D. High brachial bifurcation

96. Upper extremity venous color duplex examination reveals patency of both internal jugular and subclavian veins' however, Doppler signals are abnormally reduced. These results are indicative of which condition?

- A. Aortic stenosis
- B. Axillary vein obstruction
- ☒ C. Superior vena cava obstruction
- D. Thoracic outlet syndrome

97. Which is consistent with an abnormal digital photoplethysmography waveform?

- A. High amplitude
- B. Rapid upstroke
- ☒ C. Absent dicrotic notch
- D. Downstroke bowing toward baseline

98. Which measurement should be obtained when performing a vein mapping exam?

- ☒ A. Anterior to posterior diameter
- B. Sagittal length
- C. Peak systolic velocity
- D. Volume flow

99. Which celiac artery peak systolic velocity suggests a stenosis greater than 70%?

- A. 100 cm/s
- B. 125 cm/s
- C. 175 cm/s
- ☒ D. 225 cm/s

100. Which ankle-brachial index is most consistent with the symptom of ischemic rest pain?

- ☒ A. 0.3
- B. 0.5
- C. 0.8
- D. 1

101. In a patient with median arcuate ligament syndrome (MALS), how does intermittent compression of a visceral artery likely change the peak systolic velocities?

- A. Increase in the celiac artery with maximal inspiration
- ☒ B. Decrease in the celiac artery with maximal inspiration
- C. Increase in the superior mesenteric artery with maximal inspiration
- D. Decrease in the superior mesenteric artery with maximal inspiration

102. Which clinical finding is a contraindication to performing treadmill exercise testing?

- A. Normal ankle-brachial index at rest
- B. Thigh claudication
- ☒ C. Systolic blood pressure >180 mmHg
- D. Presence of a carotid bruit

103. A decrease in ankle pressure was noted post treadmill test, and ankle pressure returned to normal after four minutes. Which condition is most consistent with these findings?

- A. Noncompressible vessels
- ☒ B. Single-level disease
- C. Foot ischemia
- D. Normal finding

104. Which clinical condition is the most likely explanation for finding an abnormally long acceleration time (200 m/s) in the common femoral artery during arterial duplex exam of the lower extremity?

- A. Abdominal aortic aneurysm
- ☒ B. External iliac artery stenosis
- C. Common femoral artery dissection
- D. Superficial femoral artery occlusion

105. After the patient performs adequate dorsiflexions, the venous refill time measured by photoplethysmography (PPG) is greater than 20 seconds. What does this indicate?

- A. Incompetent saphenofemoral junction
- B. Incompetent perforators
- C. Incompetent saphenopopliteal junction
- ☒ D. Normal study

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- ☒ C. Measure toe pressure
- D. Place the cuffs at calf level

107. Which finding is associated with an intracranial collateral pathway in the presence of internal carotid artery occlusion?

- A. Ipsilateral retrograde anterior cerebral artery
- ☒ B. Contralateral retrograde external carotid artery
- C. Ipsilateral retrograde vertebral artery
- D. Contralateral retrograde ophthalmic artery

108. Occlusion of which carotid artery is associated with a significant increase in diastolic flow in the left external carotid artery?

- A. Right internal
- B. Right external
- ☒ C. Left internal
- D. Left common

109. Which physical finding is most consistent with retrograde arterial flow distal to a brachiocephalic arteriovenous fistula?

- ☒ A. Pallor of the hand

- B. No thrill over fistula
- C. Palpable lump or fistula
- D. Absent palpable brachial artery pulse

110. A 30-year-old woman with severe hypertension has elevated velocity in the mid renal arteries bilaterally. What is the most likely etiology?

- A. Essential hypertension
- B. Atherosclerotic renal artery stenosis
- ☒ C. Renal artery fibromuscular dysplasia
- D. Renal artery embolus

111. The profunda and genicular collateral network serve to minimize the hemodynamic abnormality caused by occlusion of which artery?

- A. Aorta
- ☒ B. Superficial femoral
- C. Iliac
- D. Hypogastric

112. Which criteria is objectively used to identify a hemodynamically significant stenosis in the upper extremity?

- A. High-resistance waveform
- B. Doppler signal beginning to alias
- ☒ C. Doubling of the peak systolic velocity
- D. Visible calcified plaque

113. Which condition is most likely to cause decreased femoral pulses and an abnormal bruit in an 18-year-old hypertensive male?

- A. Fibromuscular dysplasia
- B. Arteriovenous fistula
- C. Neurovascular compression syndrome
- ☒ D. Coarctation of the aorta

114. Which vessel is primarily used to locate the vertebral artery during an extracranial Doppler examination?

- ☒ A. Common carotid artery
- B. Internal carotid artery
- C. External carotid artery
- D. Internal jugular vein

115. During a lower extremity arterial duplex exam, circumferential calcification is noted in all tibial arteries. Which statement should be included on the preliminary report?

- ☒ A. Tibial artery assessment is limited secondary to acoustic shadowing
- B. Tibial artery assessment reveals normal flow
- C. Tibial artery assessment reveals stenosis

D. Tibial artery assessment suggests occlusion

116. Which finding is characteristic of a hemodynamically significant stenosis in the popliteal artery?

- A. Velocity increase of 200% from popliteal artery to tibioperoneal trunk
- B. High-resistive waveform in the tibioperoneal trunk
- C. Dampened systolic peak in the distal femoral artery
- ☒ D. Velocity increase of 100% from the distal femoral artery to popliteal artery

117. Which noninvasive vascular exam should be performed first when a patient presents with a history of dizziness associated with left arm exercise?

- ☒ A. Vertebral artery exam
- B. Thoracic outlet evaluation
- C. Forearm and digital pressures
- D. Upper extremity venous duplex

118. Which of the following is the main advantage of pulse recording over continuous wave Doppler for the evaluation of disease of lower extremity arteries?

- A. Allows to distinguish between occlusion and stenosis in the limb segment
- B. Provides information about the compliance of vessels in the limb segment
- C. Provides direct information of overall perfusion of a limb segment
- ☒ D. Provides indirect information of overall perfusion of a limb segment (including collaterals)

119. A patient is referred for a carotid duplex exam. No hemodynamically significant stenosis is identified. Angiography reveals the presence of a right internal carotid (ICA occlusion). What is the most likely explanation for this false negative study?

- A. A contralateral stenosis masked the ICA occlusion
- B. A hyperemic venous signal was misinterpreted as an ICA signal
- C. Color flow duplex study is insensitive to occlusion
- ☒ D. An external carotid branch was incorrectly identified as the ICA

120. When a segmental pressure examination is performed, which outcome is most likely to result from using a thigh cuff that is too narrow?

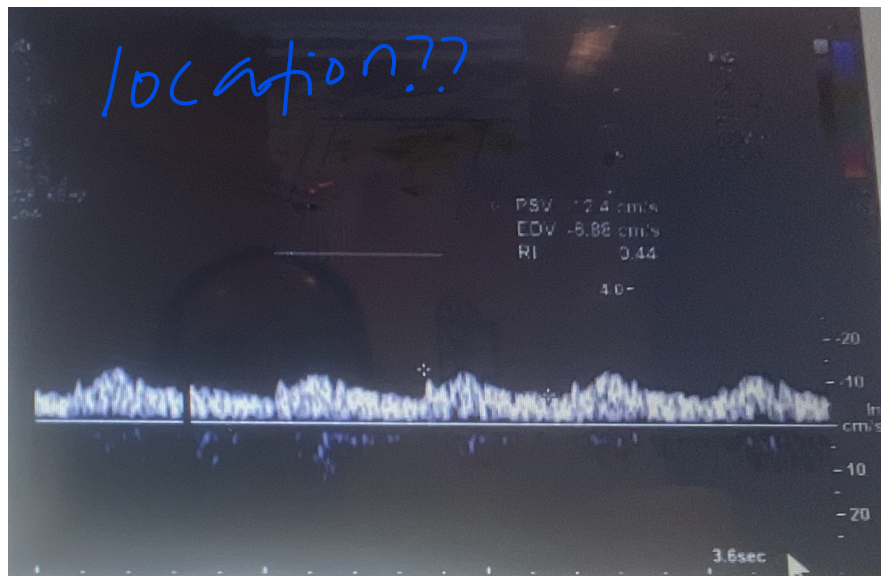
- A. Thigh pressure measurement 40 mmHg lower than brachial pressure
- ☒ B. Thigh pressure measurement 40 mmHg greater than brachial pressure
- C. Thigh pressure measurement 20 mmHg greater than brachial pressure
- D. Thigh pressure measurement 20 mmHg lower than brachial pressure

121. Following left femoral artery catheterization, ultrasound evaluation confirmed the presence of a pseudoaneurysm. Which of the following is a contraindication to ultrasound guided compression of this pseudoaneurysm?

- A. Diameter of less than 3 cm
- B. A pedicle connecting to the arterial lumen
- ☒ C. Evidence of infection

D. Duration of less than two weeks

122. Based on this image, which artery is most likely stenotic or occluded?



we
cannot
see
where
we are?
image = 😞

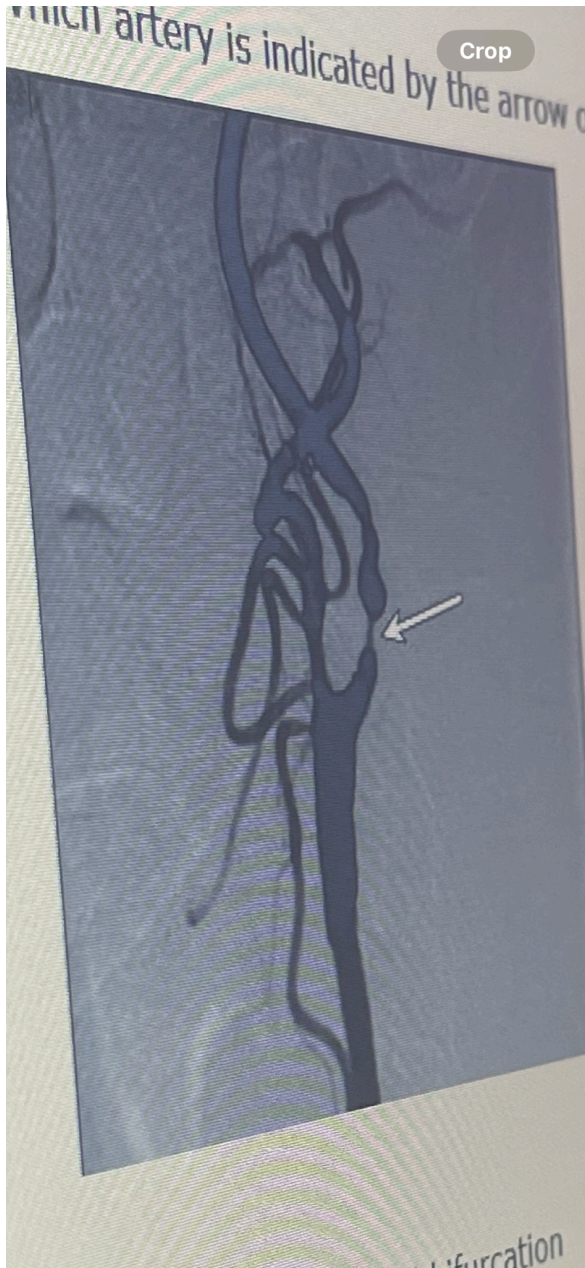
- A. Basilar
- B. Ipsilateral subclavian
- C. Contralateral vertebral
- D. Contralateral subclavian

123. Which renal artery condition is most likely represented in this image?



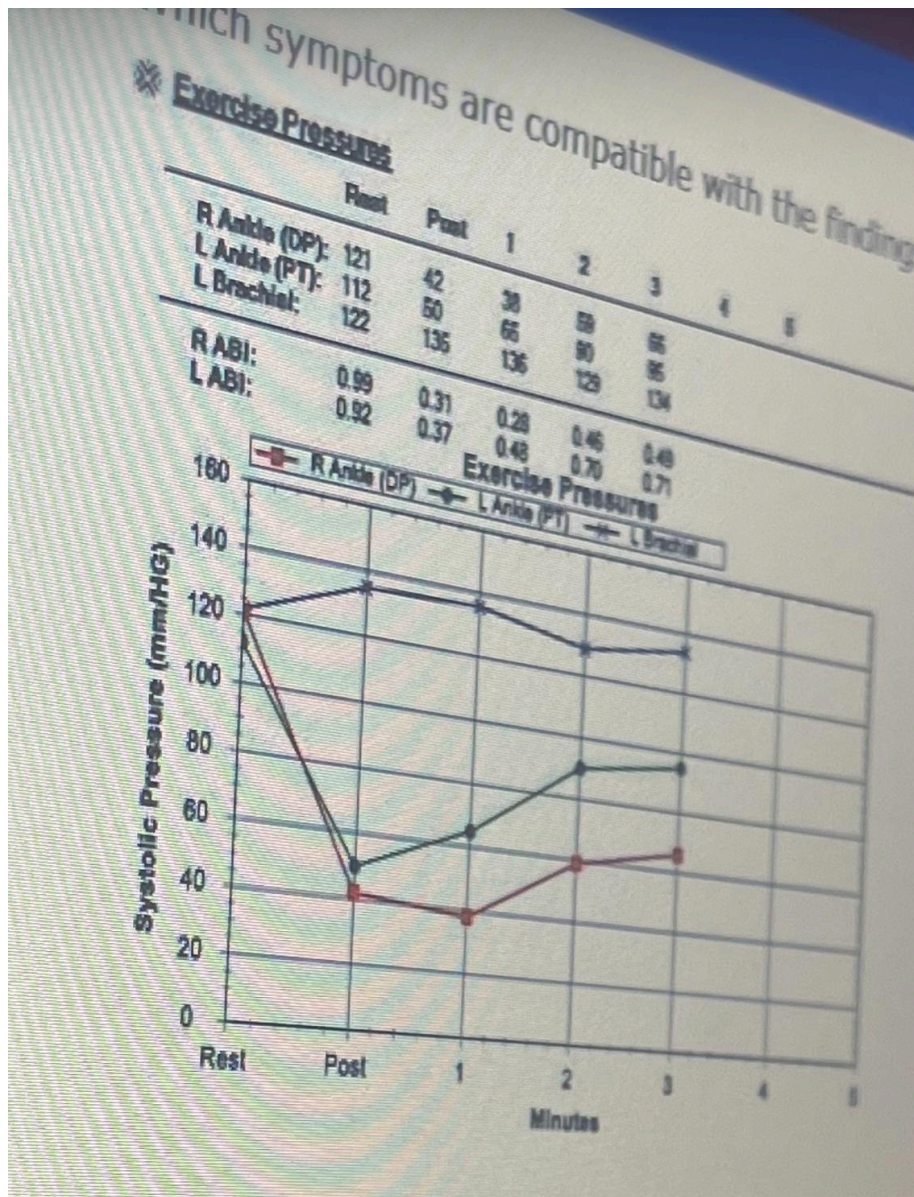
- ☒ A. Occlusion
- B. Dissection
- C. Aneurysm
- D. Fibromuscular dysplasia

124. Which artery is indicated by the arrow on this image?



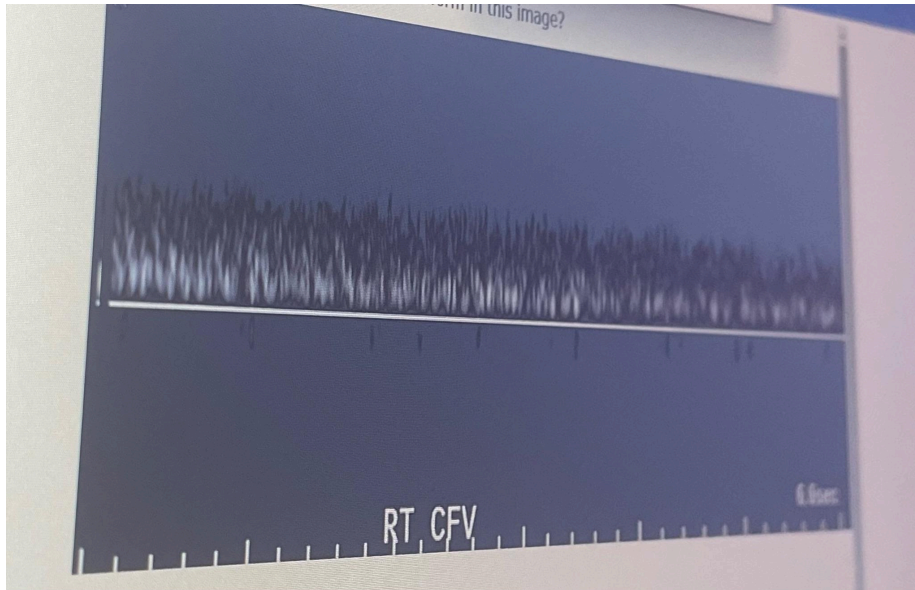
- A. Common carotid bifurcation
- ☒ B. Proximal internal carotid
- C. Distal common carotid
- D. Proximal external carotid

125. Which symptoms are compatible with findings in this image?



- A. Asymptomatic
- B. Foot pain at rest
- ☒ C. Calf pain with walking
- D. Ulcer/gangrene of the forefoot

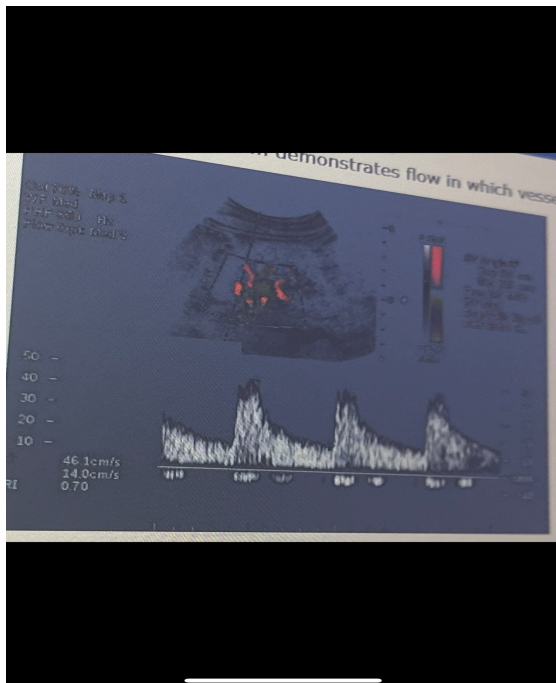
126. Which finding is indicated by the waveform in this image?

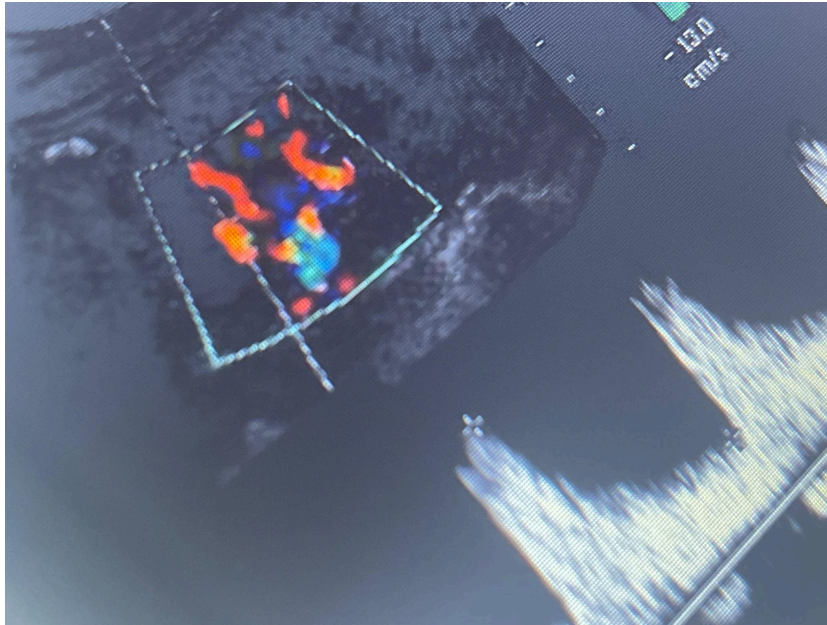


- A. Normal venous flow
- B. Downstream obstruction
- C. Congestive heart failure

☒ D. _____ stream obstruction (up? Cut off) *upstream?*

127. The spectral waveform demonstrates flow in which vessel?





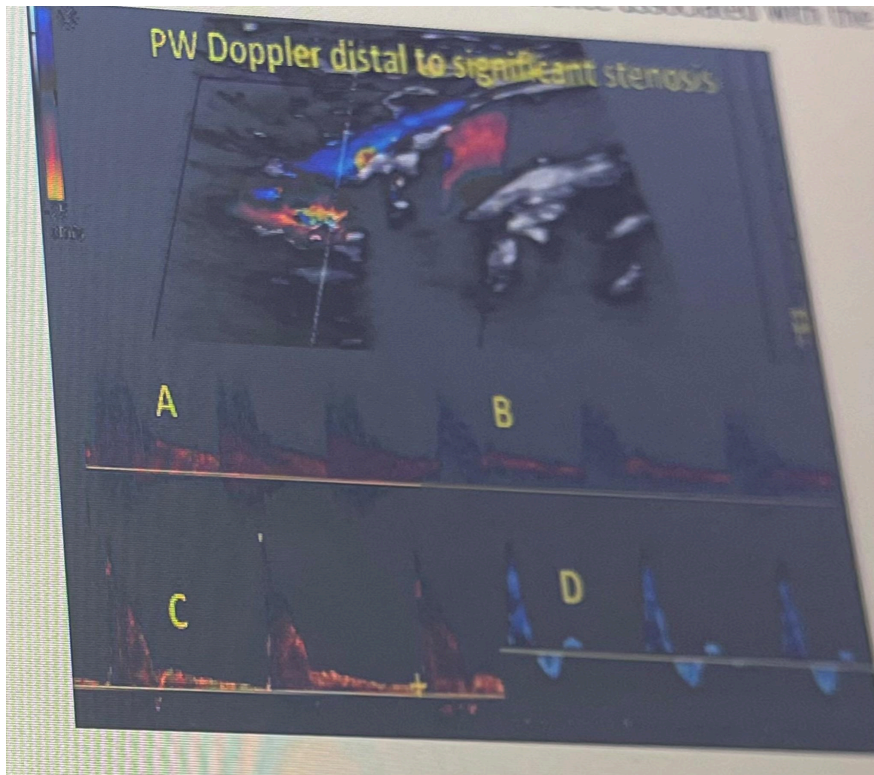
- ☒ A. Segmental artery
- ☐ B. Interlobular artery
- ☐ C. Renal vein
- ☐ D. Renal artery

128. Based on this worksheet, which artery/arteries on the left side demonstrate(s) a stenosis?

Upper Arterial			
Segmental BP			
Right		Left	
Brachial:	135	Index	95
Forearm:	125	0.93	80
Wrist (RA):	115	0.85	75
Wrist (UA):	120	0.89	70

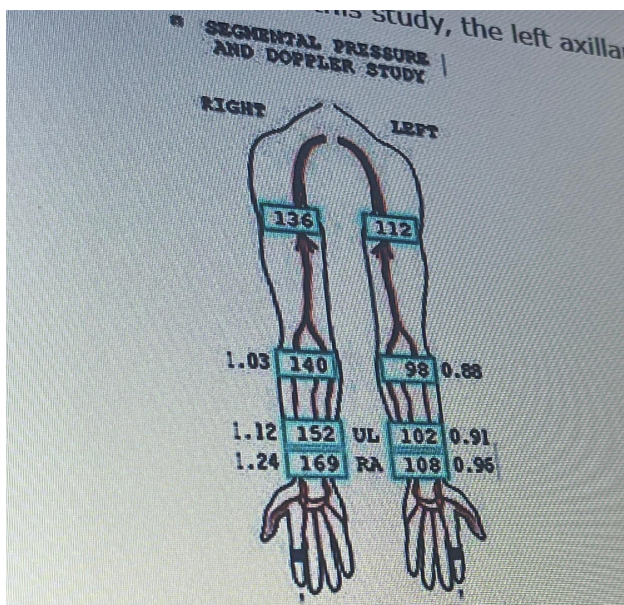
- ☐ A. Proximal radial
- ☐ B. Distal brachial
- ☐ C. Radial and ulnar
- ☒ D. Distal subclavian

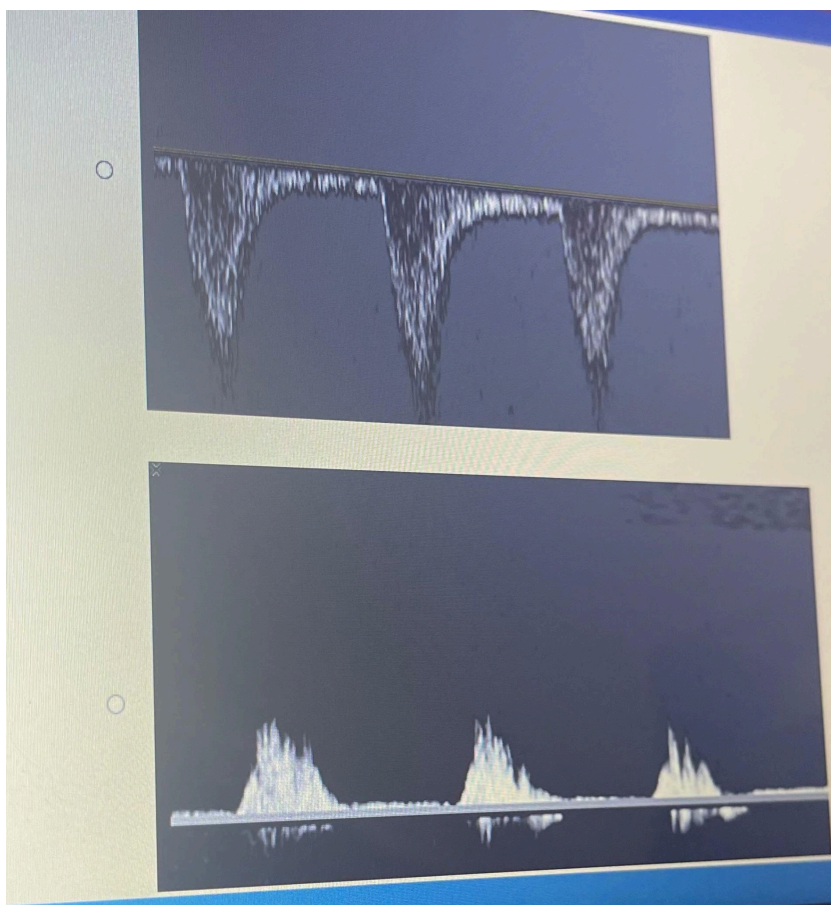
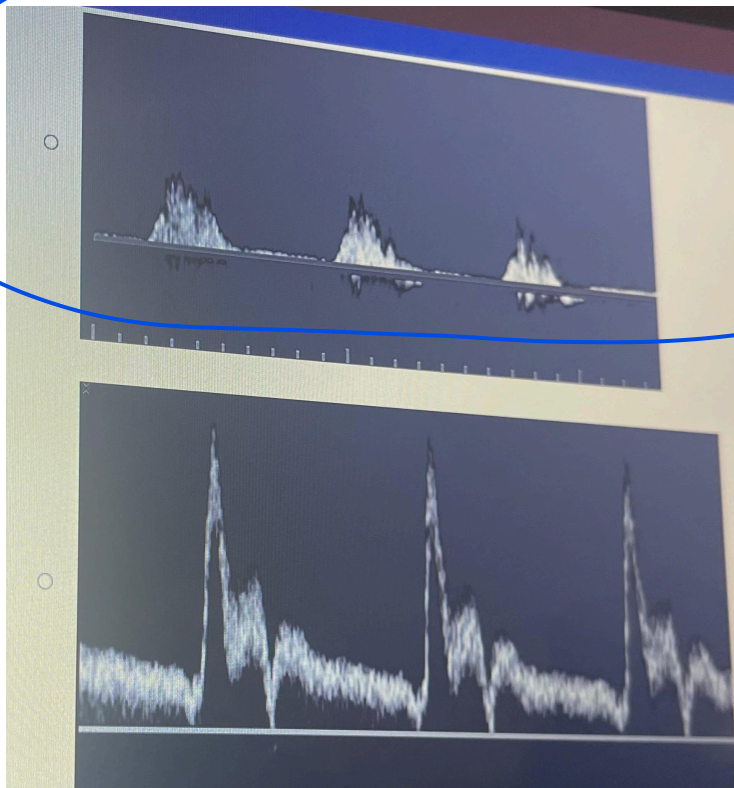
129. Which waveform represents the turbulence associated with the stenosis in this image?



- A. A
- B. B
- C. C
- D. D

130. Based on this study, the left axillary artery would demonstrate which spectral Doppler waveform?

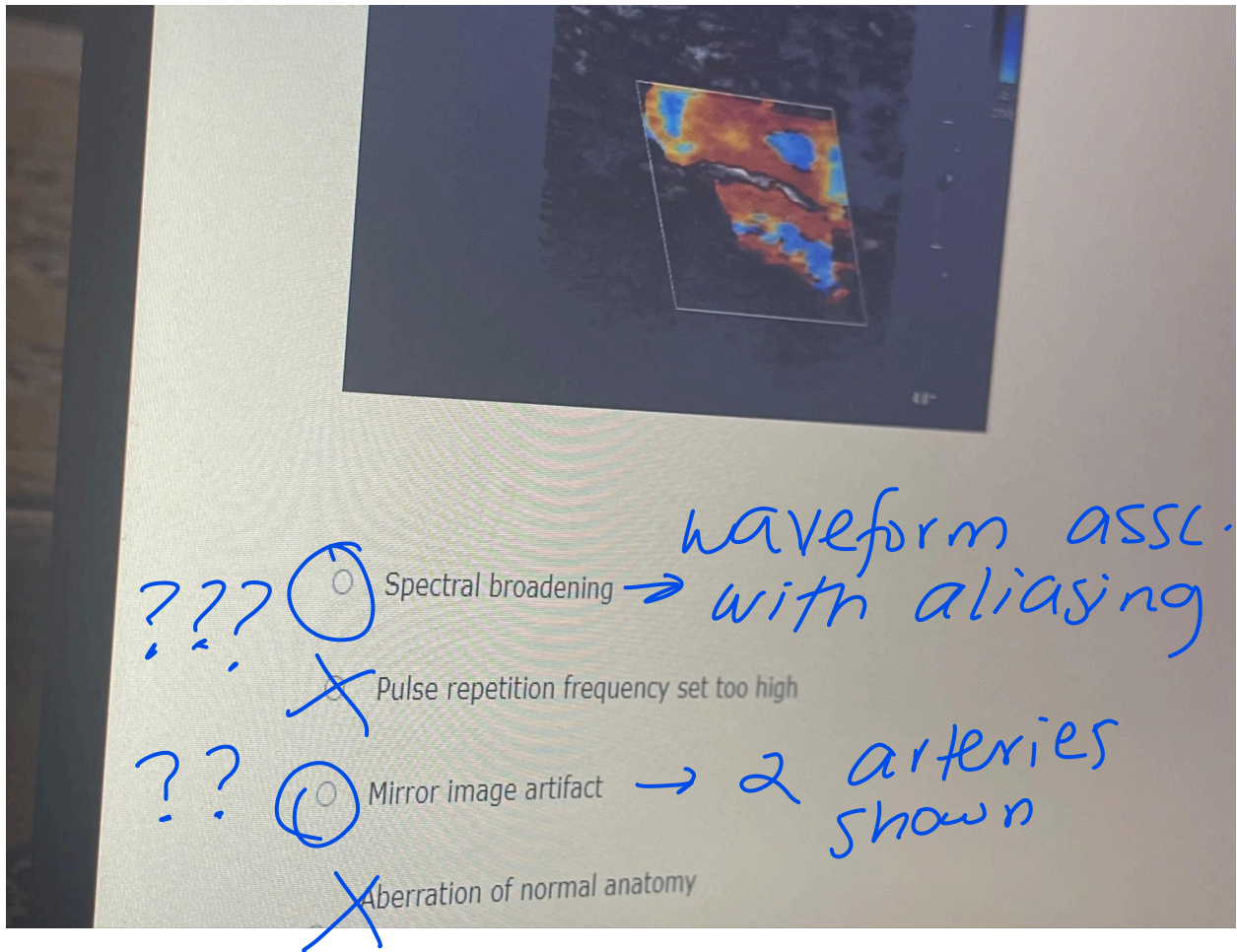




same
waveform

131.

no question?



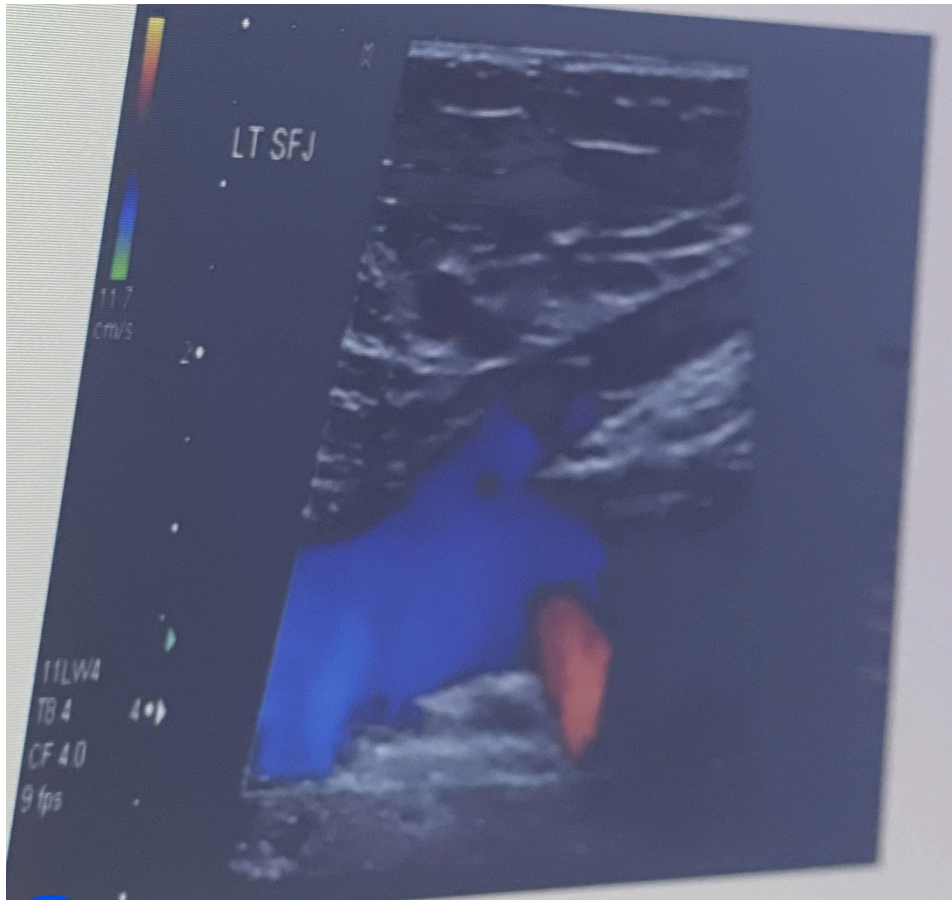
132. Which symptom is associated with occlusion of the vessel identified by the arrow on this



image?

- A. Aphasia
- B. Paresthesia
- ☒ C. Diplopia
- D. Hemiparesis

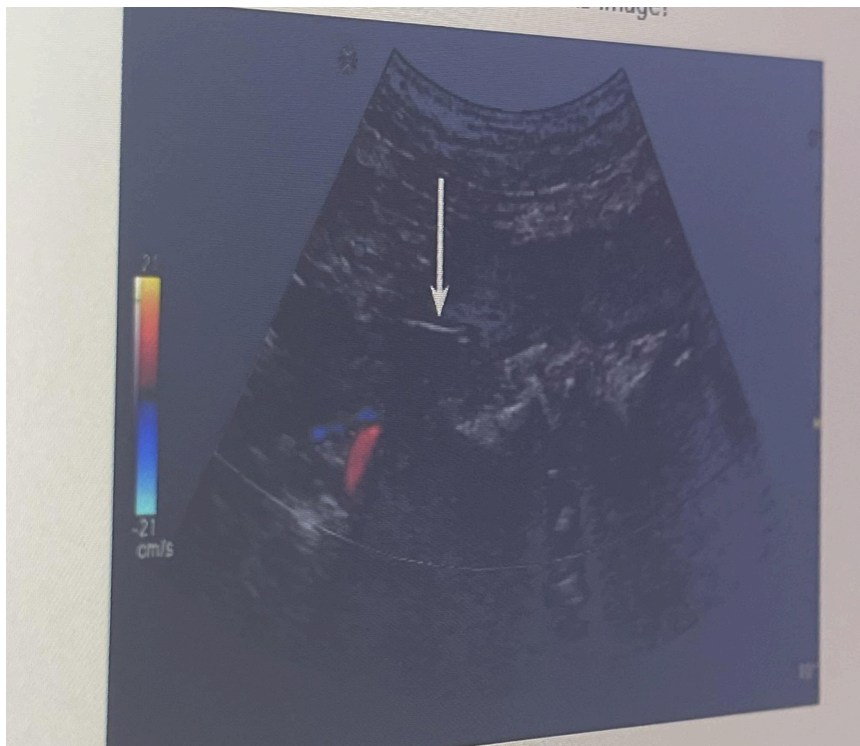
133. Which recent history is most compatible with this image?



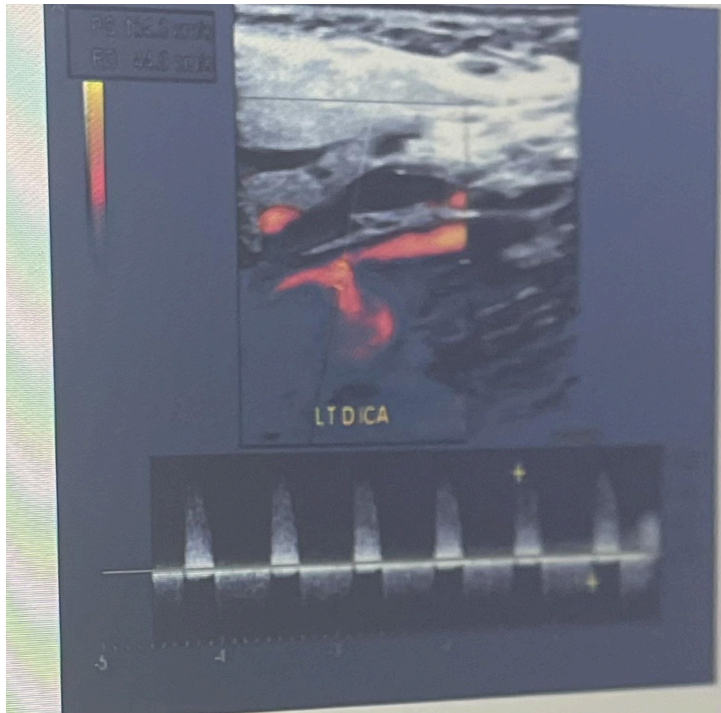
- ☒ A. Deep vein thrombosis
- ☐ B. Cardiac catheterization
- ☐ C. Vein stripping surgery
- ☐ D. Treatment for venous insufficiency

134. An abdominal ultrasound is performed on a patient with a history of deep vein thrombosis. Which finding is indicated by the arrow on this image?

- ☐ A. Bowel gas
- ☐ B. Dissection
- ☐ C. Displaced stent
- ☒ D. Inferior vena cava filter

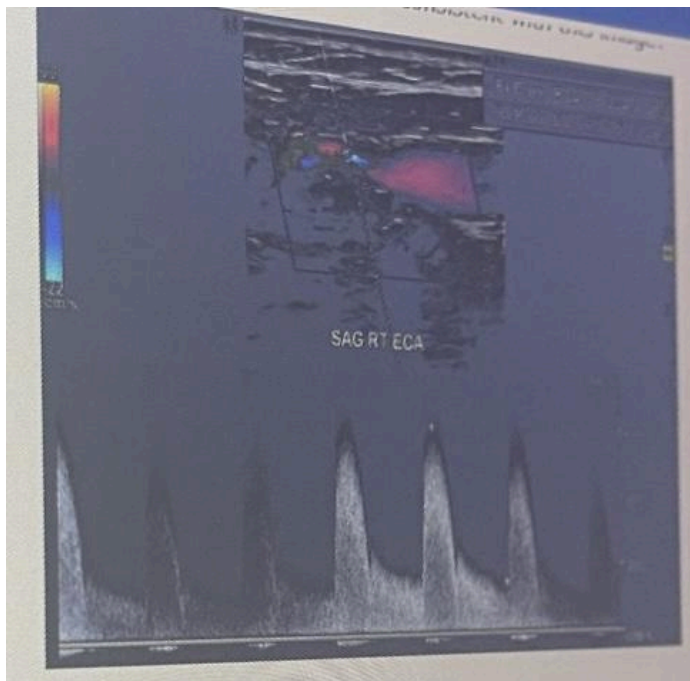


135. A patient presents for post-dissection carotid artery imaging. Which abnormality is demonstrated by this image?



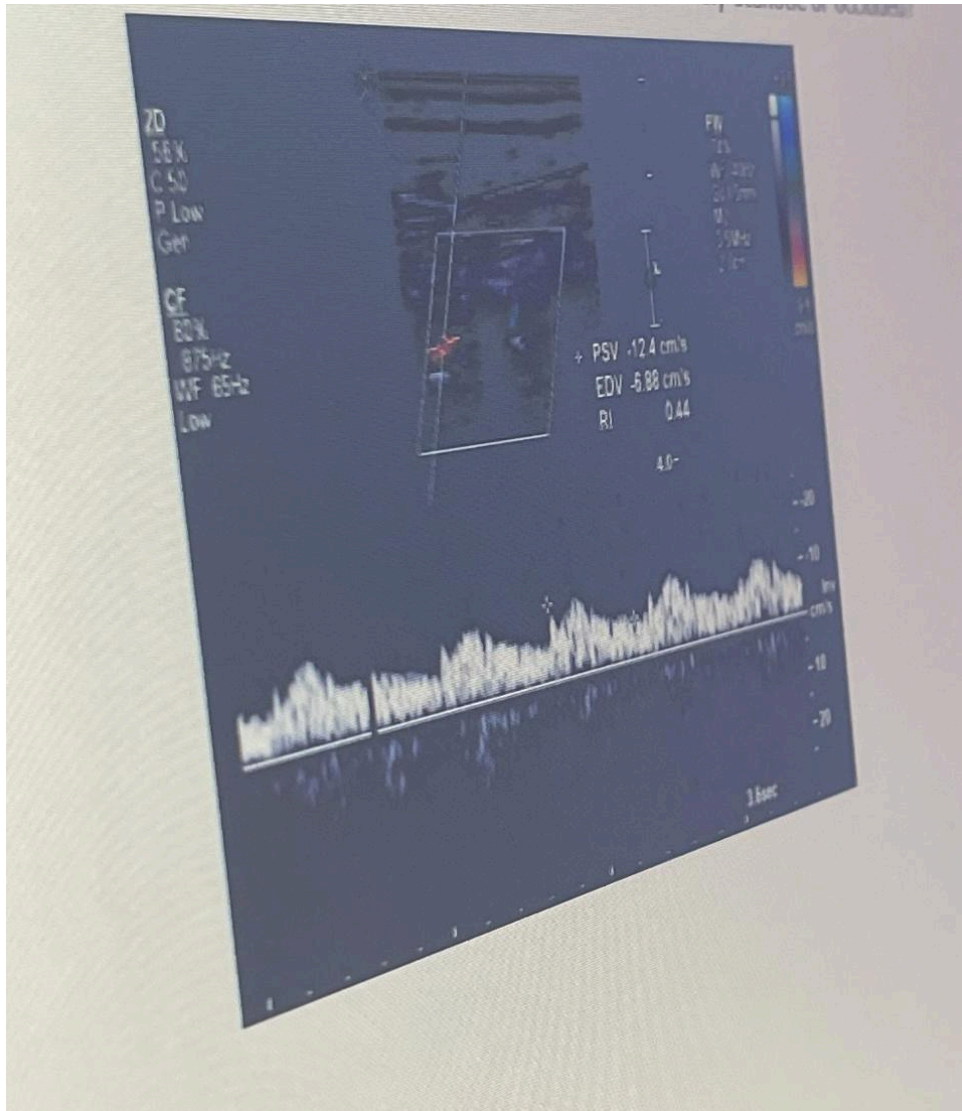
- A. Stenosis
- ☒ B. Pseudoaneurysm
- C. Arteriovenous fistula
- D. Distal internal carotid occlusion

136. Which physical finding is most consistent with this image?



- A. Arm weakness
- B. Cervical bruit
- C. Slurred speech
- ☒ D. Facial asymmetry

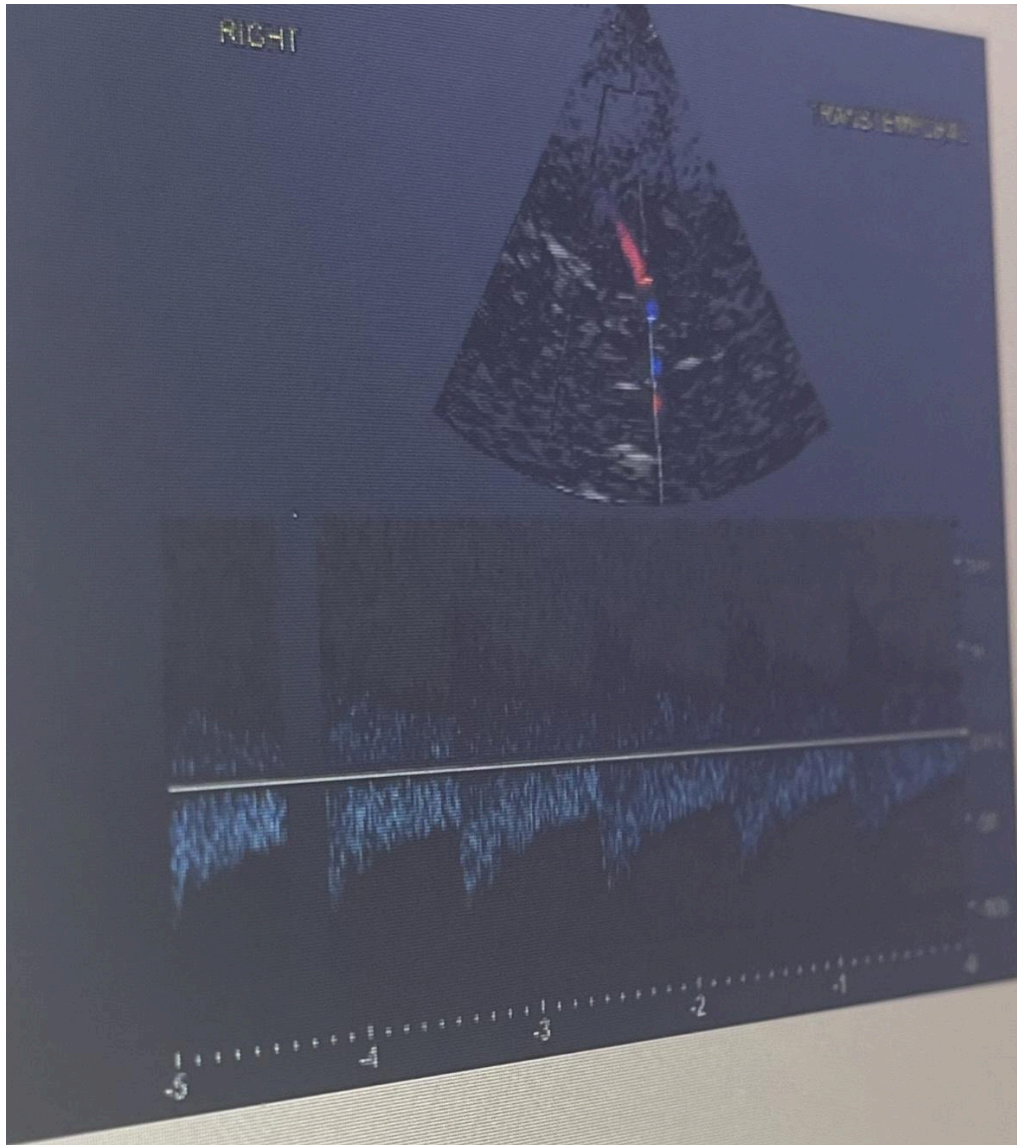
137. Based on this image, which artery is most likely stenotic or occluded?



- A. Basilar
- ☒ B. Ipsilateral subclavian — assuming this is ipsilateral vertebral?
- C. Contralateral vertebral
- D. Contralateral subclavian

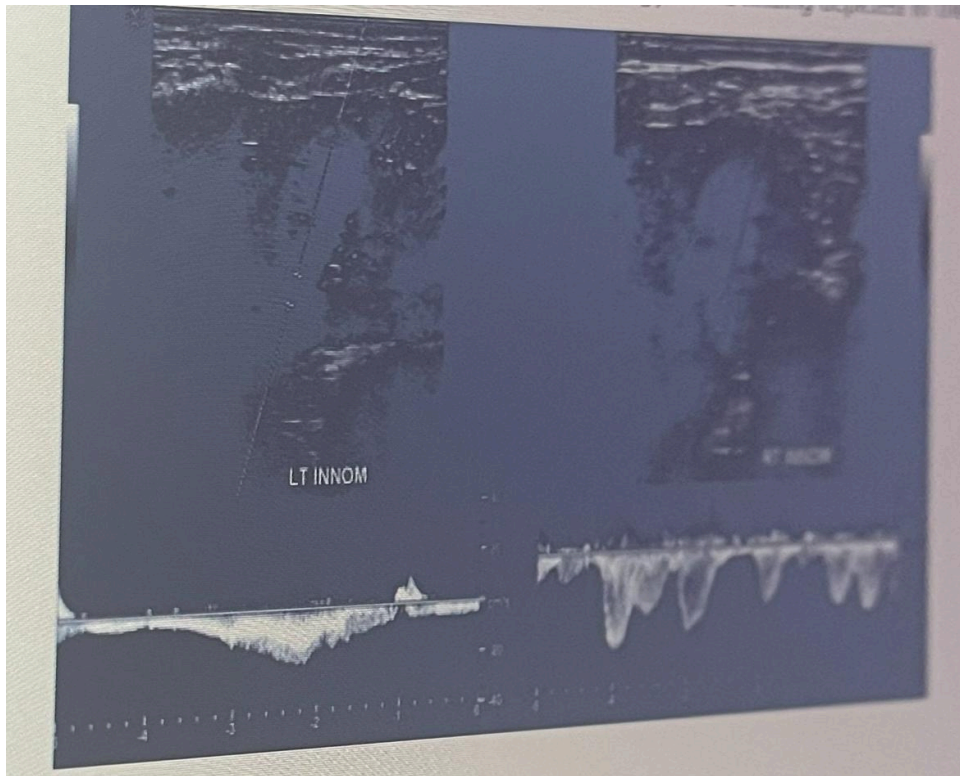
- A.

139. Which arteries are demonstrated above and below the baseline in this image?



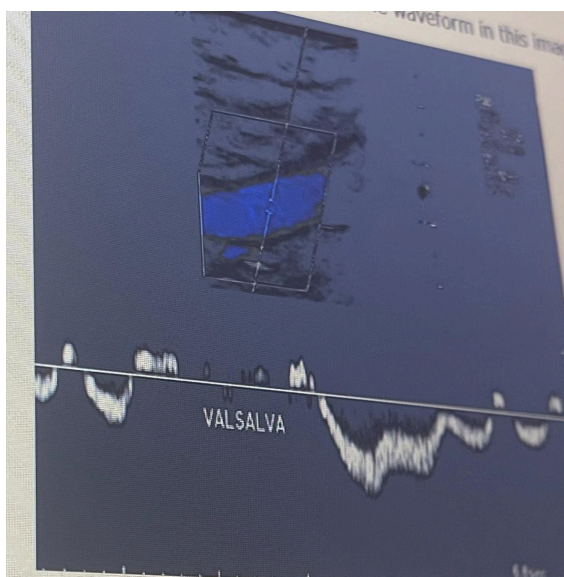
- A. Right and left middle cerebral
- B. Right and left anterior cerebral
- C. Right middle cerebral and right anterior cerebral
- D. Right anterior cerebral and right posterior cerebral

140. Which of the following best describes the etiology of the finding depicted in these images?



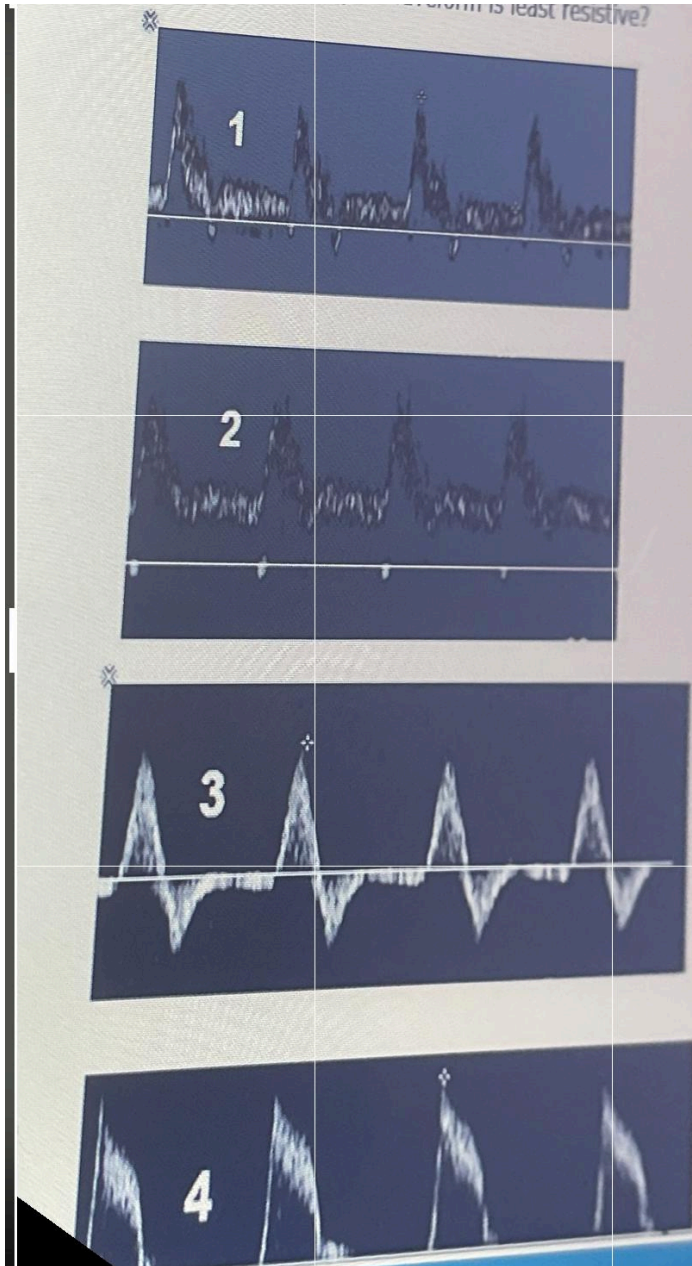
- A. Arteriovenous fistula
- B. Congestive heart failure
- ☒ C. Proximal vein obstruction
- D. Axillary thrombosis

141. Which finding is demonstrated by the waveform in this image?



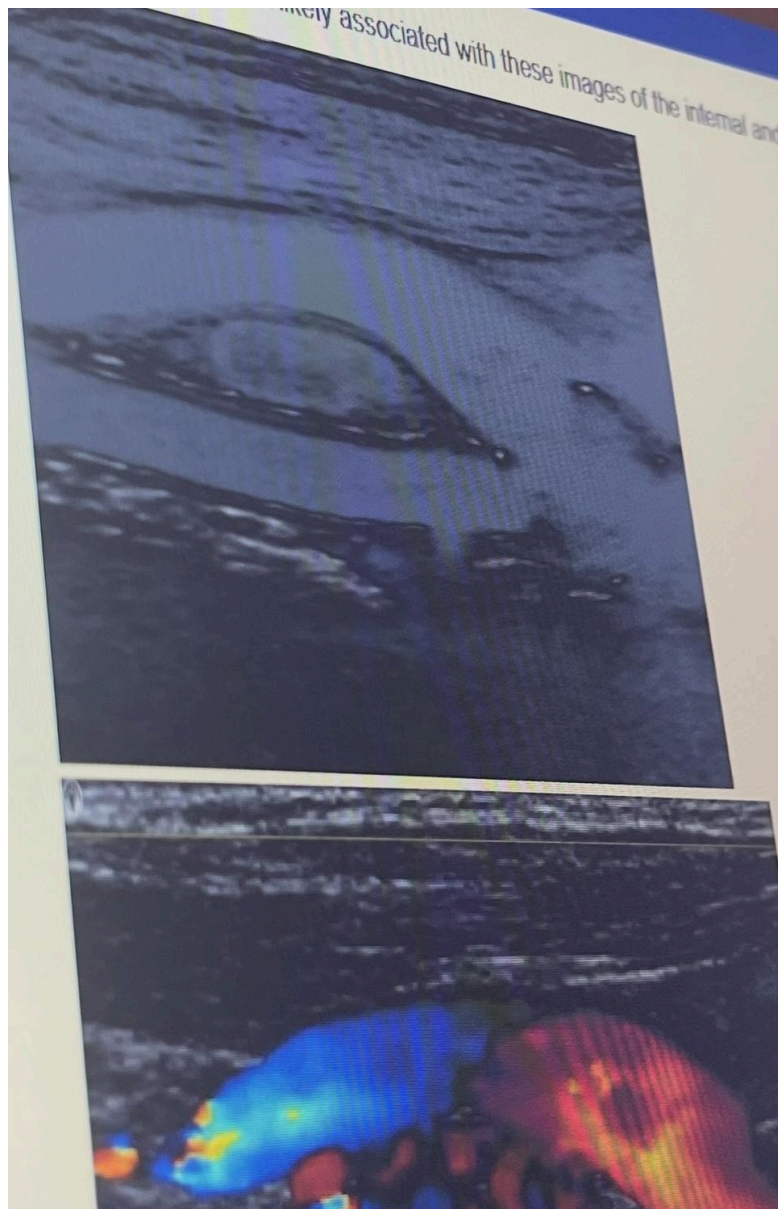
- A. Venous reflux
- ☒ B. Normal venous flow
- C. Upstream obstruction
- D. Downstream obstruction

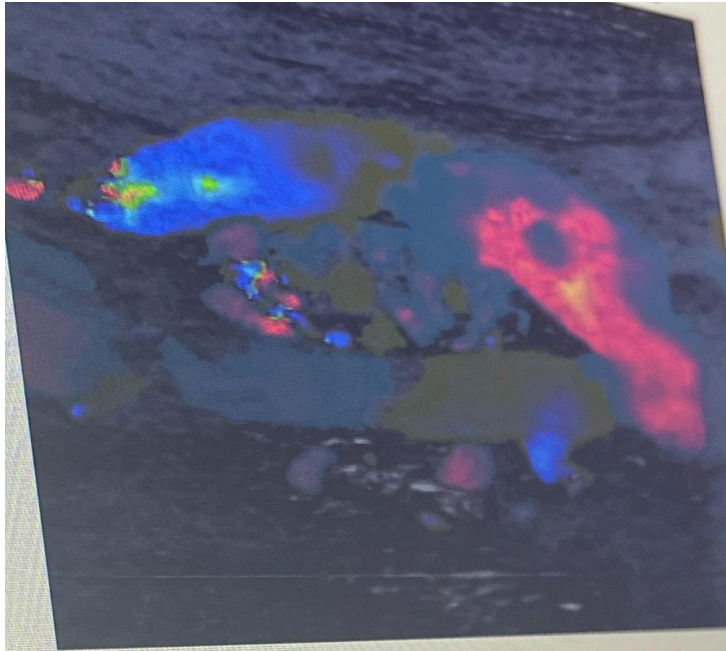
142. Which spectral Doppler waveform is least resistive?



- A. 1
- ☒ B. 2
- C. 3
- D. 4

143. Which condition is most likely associated with these images of the internal and external carotid arteries?





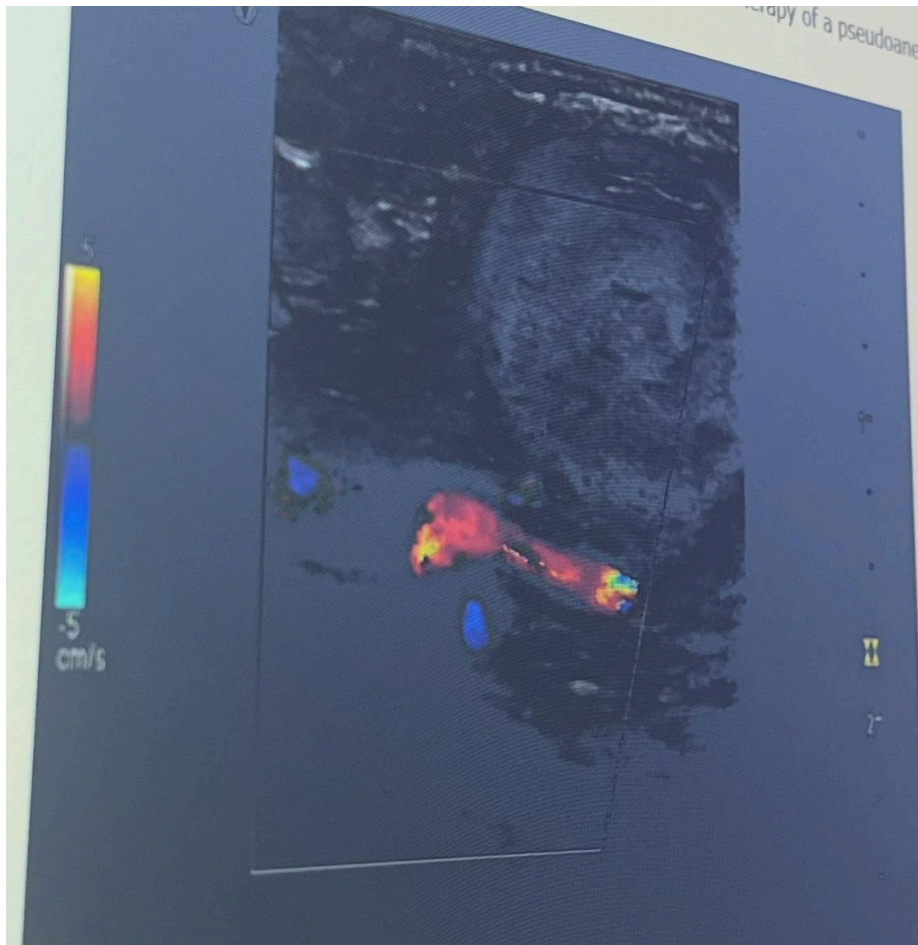
- A. Normal
- B. Enlarged thyroid
- C. Carotid body tumor
- ☒ D. Enlarged lymph node

144. Which artifact is displayed in this image?



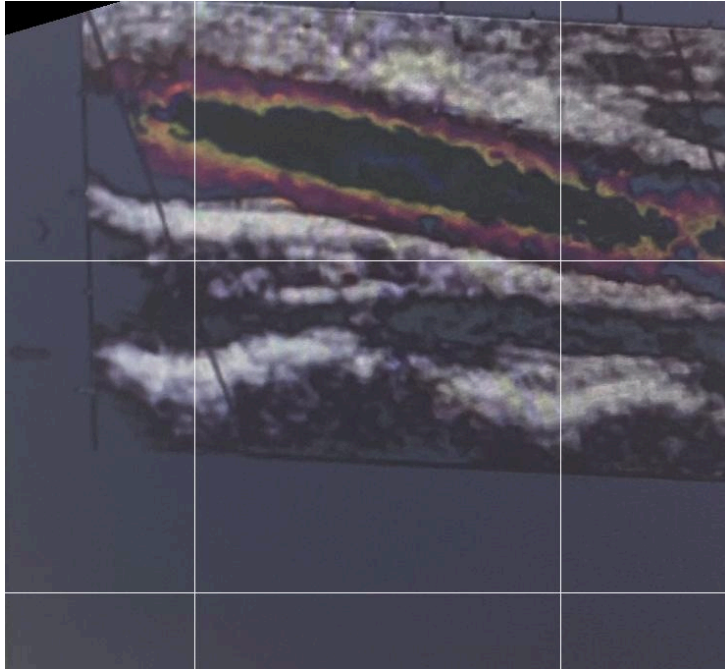
- A. Comet tail
- B. Reverberation
- C. Enhancement
- ☒ D. Acoustic shadowing

145. This image was taken during transcatheter compression therapy of a pseudoaneurysm. What is the next course of action?



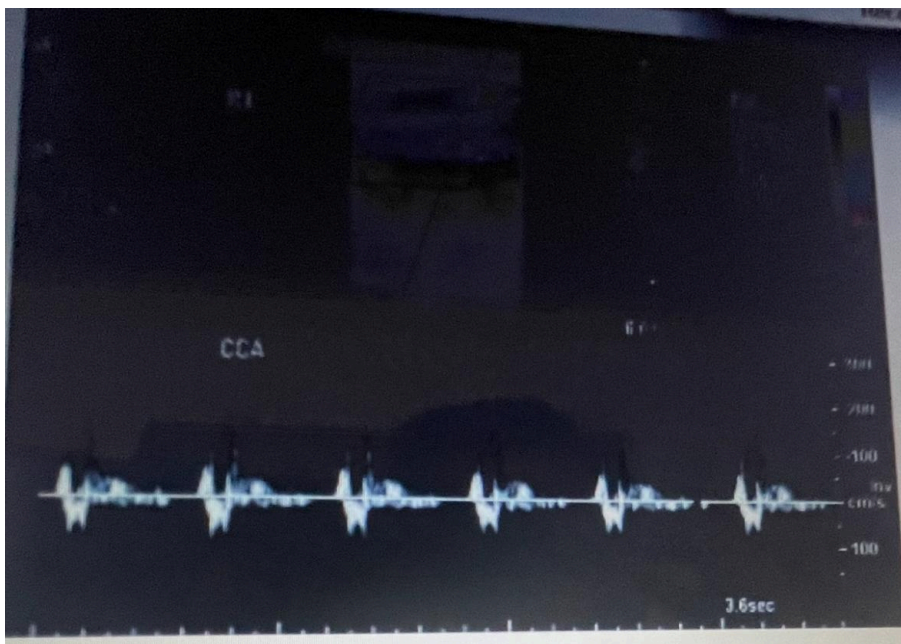
- ☒ A. Procedure complete
- ☐ B. Further compression required
- ☐ C. Surgical therapy required
- ☐ D. Failure of therapy

146. Which artifact is displayed in this image?



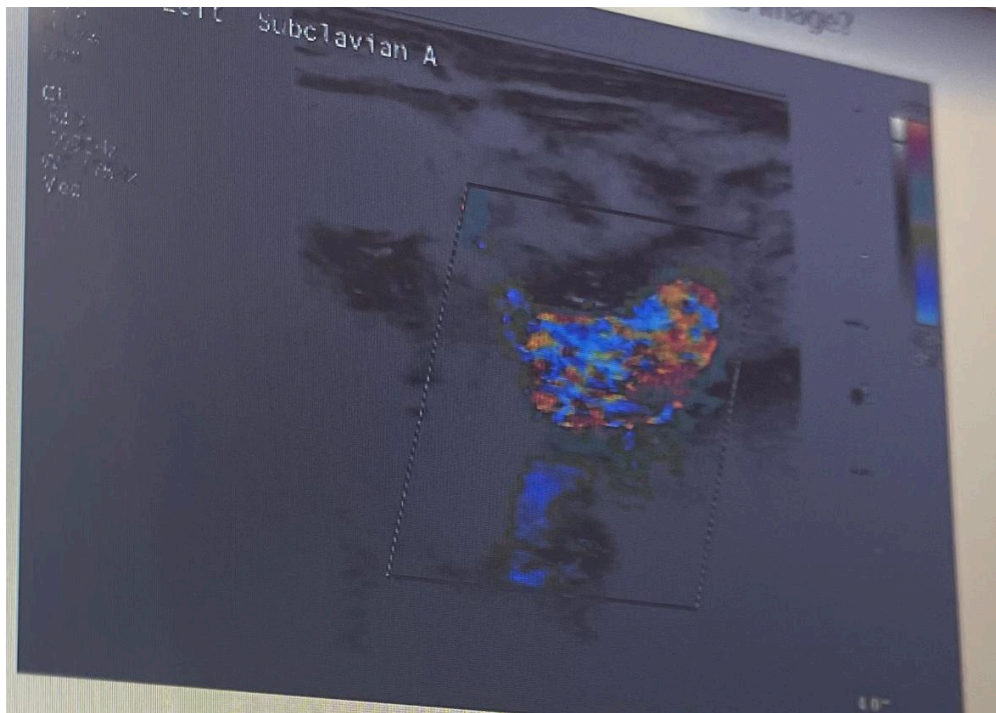
- A. Color flash
- ☒ B. Aliasing
- C. Side-lobe
- D. ?

147.



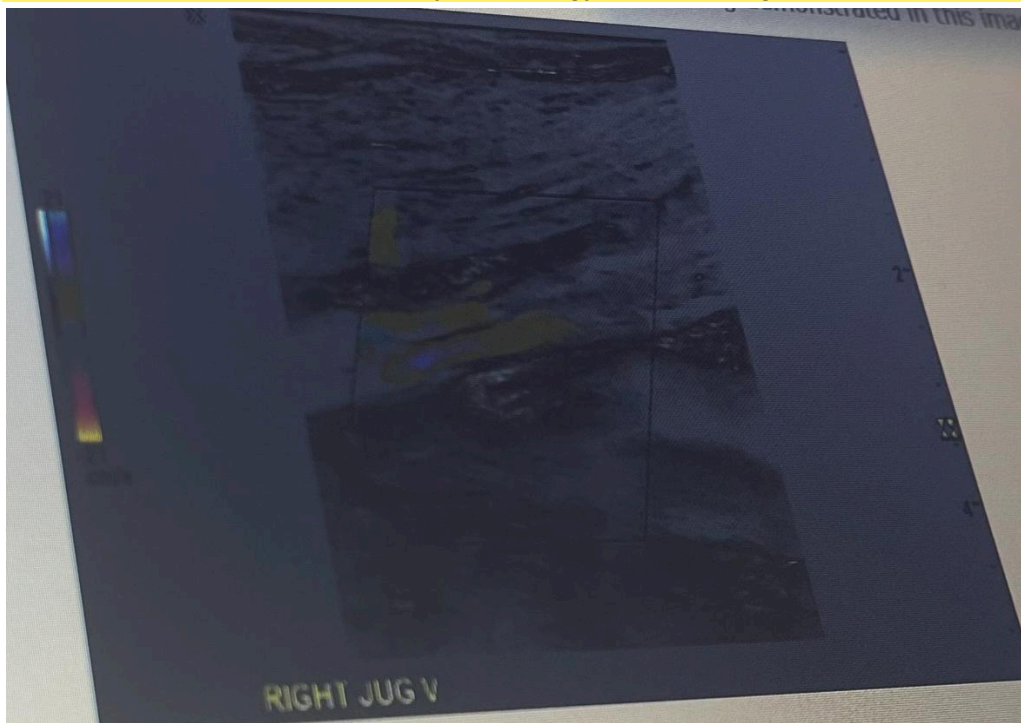
- A. Dissection
- B. Stenosis
- ☒ C. Aneurysm
- D. Arteriovenous fistula

148. Which change should be made to optimize this image?



- ☒ A. Increase color scale
- ☐ A. Decrease color scale
- ☐ B. Increase color gain
- ☐ C. Decrease overall gain

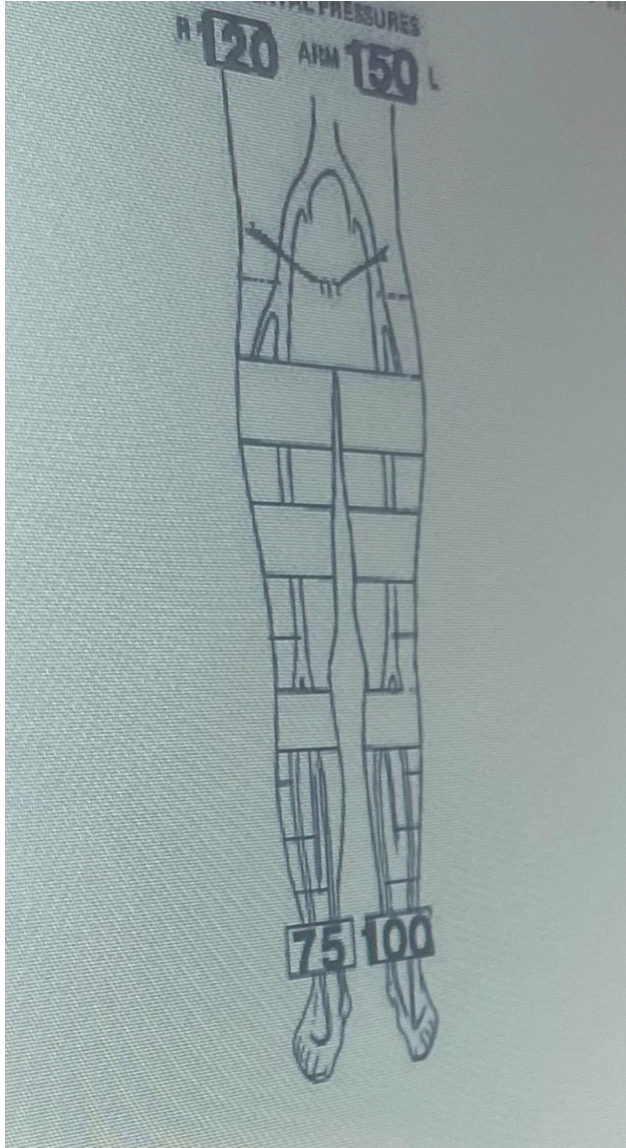
149. Which condition is most likely an etiology of the finding demonstrated in this image?



- A. Acute cervical trauma
- B. Thoracic outlet syndrome
- C. Costoclavicular space narrowing
- ☒ D. History of central venous catheter

??

150. Following the pressures given in this image, what is the ankle-brachial index _____?



- ☒ A. 0.50
- B. 0.63
- C. 0.67
- D. 0.83

? we don't get
calculator on exam.

